Graph methods for imaging, Vision and computing (B31RX) 2025/26

Tutorial 4: Gaussian mixture model (GMM) clustering and the Expectation-Maximization algorithm

In this tutorial, we will investigate a clustering method relying on a Gaussian mixture model (GMM) and an algorithm to train the corresponding classifier, namely the Expectation-Maximization algorithm.

In contrast to K-Means, the GMM clustering algorithm assumes that the clusters have parametric shapes, i.e., (multivariate) Gaussian distributions. This algorithm has a Bayesian interpretation and learns the cluster parameters by maximum marginal likelihood estimation (MMLE) or marginal maximum a posteriori (MMAP) estimation. In this tutorial, we will use MMLE estimation. Thanks to its Bayesian formulation, it provides not only estimated labels for each sample, but also membership probabilities, i.e., for each sample, the probabilities of belonging to each cluster.

Moreover, this clustering is appropriate when the different clusters have: 1) different shapes and 2) different numbers of samples per class.

Background

<u>Data model:</u>

Assume we have N observations $\boldsymbol{y} = \{\boldsymbol{y}_1, \ \dots, \ \boldsymbol{y}_N\}$, assumed to be i.i.d., that we would like to cluster into K classes. The clusters are assumed to be Gaussian. More precisely, the cluster k is characterized by its mean $\boldsymbol{\mu}_k$ and its covariance matrix $\boldsymbol{\Sigma}_{\mathrm{k}}$. The probability (for each sample) of belonging to the cluster k is π_k. Since each sample is assumed to belong to a single cluster, we have

$$\sum_{k=1}^{K} \pi_k = 1.$$

The cluster parameters are gathered in $\boldsymbol{\theta}$ such that $\boldsymbol{\theta} = \{\boldsymbol{\mu}_{\boldsymbol{k}}, \boldsymbol{\Sigma}_{\mathbf{k}}, \pi_k\}_k$.

Each data point $\boldsymbol{y}_n$ is associated with a discrete label denoted $l_n \in \{1, \dots, K\}$ and to simplify the derivation of the EM algorithm, we introduce $\boldsymbol{x} = \{l_1, \ \dots, \ l_N\}$.

For each data, the generative model can be expressed as

$$\boldsymbol{y}_n | (l_n = k, \ \boldsymbol{\theta}) \sim \boldsymbol{\mathcal{N}}(\boldsymbol{\mu}_{\boldsymbol{k}}, \boldsymbol{\Sigma}_{\mathbf{k}}),$$

which essentially states that, assuming that if the nth data point belongs to cluster k (i.e.,$l_n = k$), and given the cluster parameters in $\boldsymbol{\theta}$, $\boldsymbol{y}_n$ follows a Gaussian distribution whose mean and covariance matrix depend on the cluster index.

Given the data in $\boldsymbol{y}$, the clustering problem is formulated as the estimation of the unknown parameters $\boldsymbol{\theta}$ and x. While point estimates are usually sufficient for $\boldsymbol{\theta}$, it would be good to extract not only assign a label to each data point, but also estimate the probability of belonging to each of the clusters (called membership probabilities). This is something that is possible using the Expectation-Maximization algorithm.

Expectation-Maximization algorithm:

The EM algorithm is a powerful algorithm that can be used when Bayesian models include so-called latent variables. Let $\boldsymbol{y} = \{\boldsymbol{y}_1, \ \dots, \ \boldsymbol{y}_N\}$ be a set of observations associated with a set of unknown variables ($\boldsymbol{\theta}$ and $\boldsymbol{x}$). We separate $\boldsymbol{\theta}$ and $\boldsymbol{x}$ as $\boldsymbol{x}$ is a set of variables referred to as latent variables.

The EM algorithm can be applied to a variety of problems but scenarios where its derivation is relatively simple include cases where

- $\boldsymbol{\theta}$ is deterministic
- $f(\boldsymbol{y} \,|\, \boldsymbol{x}, \boldsymbol{\theta})$ is available and $\boldsymbol{y}$ is observed
- $f(\boldsymbol{x}|\boldsymbol{\theta})$ is known

In this context, the EM algorithm aims to solve jointly

$$\begin{cases} \widehat{\boldsymbol{\theta}}_{MMLE} = argmax_{\boldsymbol{\theta}} f(\boldsymbol{y}|\boldsymbol{\theta}) \\ \widehat{\boldsymbol{x}}_{MMSE} = \mathbb{E}_{f(\boldsymbol{x}|\boldsymbol{y},\widehat{\boldsymbol{\theta}}_{MMLE})}[\boldsymbol{x}] \end{cases}$$

Instead of first estimating $\widehat{\boldsymbol{\theta}}_{MMLE}$ and then $\widehat{\boldsymbol{x}}_{MMSE}$ (based on $\widehat{\boldsymbol{\theta}}_{MMLE}$ estimated first), the EM algorithm will iteratively update current estimates of $\boldsymbol{\theta}$ and $\boldsymbol{x}$. It turns out that this strategy is often simpler that computing $f(\boldsymbol{y}|\boldsymbol{\theta}) = \int f(\boldsymbol{y}, \boldsymbol{x}|\boldsymbol{\theta})\mathrm{d}\boldsymbol{x}$ needed to compute $\widehat{\boldsymbol{\theta}}_{MMLE}$.

Assume at iteration t that we have an estimate of $\boldsymbol{\theta}$, denoted $\boldsymbol{\theta}^{(t)}$. Each iteration of the EM algorithm we consist of two steps 1) an Expectation step (E-Step) and then 2) a Maximization step (M-step) as follows

- E-Step: Compute

$$Q\left(\boldsymbol{\theta}, \ \boldsymbol{\theta}^{(t)}\right) = \mathbb{E}_{f(\boldsymbol{x}|\boldsymbol{y},\boldsymbol{\theta}^{(t)})}\left[\log\left(f(\boldsymbol{y}, \boldsymbol{x} \mid \boldsymbol{\theta})\right)\right]$$

- M-Step: Solve

$$\boldsymbol{\theta}^{(t+1)} = argmax_{\boldsymbol{\theta}} \ Q\left(\boldsymbol{\theta}, \ \boldsymbol{\theta}^{(t)}\right)$$

In this tutorial, we will derive the steps of the EM algorithm for GMM-based clustering. We will first derive the different update rules needed and we will then use Python libraries to perform GMM clustering on a simple example with 3D data points.

Question 1

To perform the E-Step, we first need to compute $f(\boldsymbol{y}, \boldsymbol{x} \mid \boldsymbol{\theta}) = f(\boldsymbol{y}|\boldsymbol{x}, \boldsymbol{\theta}) f(\boldsymbol{x}|\boldsymbol{\theta})$. Assuming that the data points are mutually independent given $(\boldsymbol{x}, \boldsymbol{\theta})$, what is the probability density function $f(\boldsymbol{y}|\boldsymbol{x}, \boldsymbol{\theta})$? (Hint: since we will compute the logarithm of this density, it is more convenient to express it as a product of factors instead of sums).

Question 2

Similarly, assuming that the labels in $\boldsymbol{x}$ are mutually independent conditioned on $\boldsymbol{\theta}$, what is the expression of $f(\boldsymbol{x}|\boldsymbol{\theta})$?

Question 3

Using the Bayes' rule, write down the expression of $\log\big(f(\boldsymbol{y}, \boldsymbol{x} \mid \boldsymbol{\theta})\big)$.

Question 4

To complete the E-Step, we need to compute the expectation of the expression above w.r.t. $f(\boldsymbol{x}|\boldsymbol{y}, \boldsymbol{\theta}^{(t)})$. Thus, we need to compute $f(\boldsymbol{x}|\boldsymbol{y}, \boldsymbol{\theta}^{(t)})$. Using the Bayes' rule, show that

$$f\big(\boldsymbol{x}\big|\boldsymbol{y}, \boldsymbol{\theta}^{(t)}\big) = \prod_{n=1}^{N} f(l_n|y_n, \boldsymbol{\theta}^{(t)})$$

and compute $f\big(l_n = k\big|y_n, \boldsymbol{\theta}^{(t)}\big)\ \forall (n, k)$.

To simplify notations, we will denote these probabilities by $p_{n,k} = f\big(l_n = k\big|y_n, \boldsymbol{\theta}^{(t)}\big)$.

Question 5

Compute $Q\big(\boldsymbol{\theta},\ \boldsymbol{\theta}^{(t)}\big)$, which completes the E-Step.

Question 6

By differentiating $Q\big(\boldsymbol{\theta},\ \boldsymbol{\theta}^{(t)}\big)$ w.r.t. $\boldsymbol{\pi} = \{\pi_1, \dots, \pi_K\}$ and setting the gradient to 0, find the new vector $\boldsymbol{\pi}^{(\boldsymbol{t}+\boldsymbol{1})}$.

Question 7

Similarly, by differentiating $Q\big(\boldsymbol{\theta},\ \boldsymbol{\theta}^{(t)}\big)$ w.r.t. $\boldsymbol{\mu_k}$ and setting the gradient to 0, find the new vector $\boldsymbol{\mu}_{\boldsymbol{k}}^{(\boldsymbol{t}+\boldsymbol{1})}$.

Question 8

Finally by differentiating $Q\left(\boldsymbol{\theta},\ \boldsymbol{\theta}^{(t)}\right)$ w.r.t. $\boldsymbol{\Sigma_k}$ and setting the gradient to 0, find the new covariance matrix $\boldsymbol{\Sigma_k^{(t+1)}}$.

Questions 9 – 13

See Jupyter Notebook